

MA 442 - Final exam practice

Name: _____ **BUID:** _____

There are six problems, you must solve **five** of them. Each problem is worth **10 points**.
You have 90 minutes to complete this exam.

This exam is closed-book, but you are allowed a single cheat sheet.

Question 1. For each statement below, circle whether it is **True** or **False**.

Statement	T	F
(a) If A is an $n \times n$ matrix with $\det(A) = 0$, then A has an eigenvalue 0.	☐	☐
(b) If $AB = 0$ for square matrices A, B , then either $A = 0$ or $B = 0$.	☐	☐
(c) Every linearly independent set in a finite-dimensional vector space can be extended to a basis.	☐	☐
(d) If $T: V \rightarrow V$ has $\ker(T) = \{0\}$, then T is invertible.	☐	☐
(f) If A is diagonalizable, then A^2 is diagonalizable.	☐	☐
(g) If A^2 is diagonalizable, then A is diagonalizable.	☐	☐
(h) If A, B are diagonalizable and $AB = BA$, then they are simultaneously diagonalizable.	☐	☐
(i) The determinant is a linear function of the columns of a matrix.	☐	☐
(j) If λ is an eigenvalue of T , then $T - \lambda I$ is not invertible.	☐	☐
(k) Every $n \times n$ real matrix has n real eigenvalues.	☐	☐
(l) If A is invertible, then $\det(A^{-1}) = (\det A)^{-1}$.	☐	☐
(o) If A is upper triangular, then its eigenvalues are the diagonal entries.	☐	☐
(p) Every square matrix satisfies its characteristic polynomial.	☐	☐
(q) If A satisfies $p(A) = 0$, then $p(t)$ must be a multiple of $\chi_A(t)$.	☐	☐
(r) If $U, W \subseteq V$, then $\dim(U + W) = \dim(U) + \dim(W)$.	☐	☐
(s) If $T^2 = 0$, then $\text{im}(T) \subseteq \ker(T)$.	☐	☐
(t) Any two bases of a finite-dimensional vector space have the same size.	☐	☐

Question 2. Let $T: V \rightarrow V$ be a diagonalizable linear transformation.

- (a) Define $\det T$.
- (b) Let λ be an eigenvalue of T and $E_\lambda \subset V$ its associated eigenspace. Show that E_λ is T -invariant.
- (c) Show that

$$\det(T) = \prod_{\lambda} \det(T|_{E_\lambda}). \quad (1)$$

Question 3. Let V be an n -dimensional real vector space and let $\omega \in \text{Alt}_n(V, \mathbb{R})$ be an n -linear alternating function. Let $T, S: V \rightarrow V$ be two linear transformations and assume that $v_1, \dots, v_n$ are linearly independent. Show that:

$$\omega(TSv_1, \dots, TSv_n) = \frac{\omega(Tv_1, \dots, Tv_n)\omega(Sv_1, \dots, Sv_n)}{\omega(v_1, \dots, v_n)}. \quad (2)$$

Question 4. Let $V = \mathbb{R}[x]_{\leq 2}$ and define

$$T(f)(x) = f(x+1) - f(1).$$

- (a) Compute the matrix of T with respect to the basis $\{1, x, x^2\}$.
- (b) Find the characteristic polynomial of T .
- (c) Determine the eigenvalues and eigenspaces of T .
- (d) Is T diagonalizable?

Question 5. (a) State the definition of an *isomorphism* (denoted $\cong$) between two vector spaces.

(b) Let Vect be the set of all finite dimensional vector spaces. Show that the notion of isomorphism $\cong$ is an *equivalence relation* on this set.

(c) Produce a bijection of sets

$$\delta: \text{Vect}/\cong \rightarrow \mathbb{Z}_{\geq 0} \tag{3}$$

where $\mathbb{Z}_{\geq 0}$ is the set of non-negative integers.

Question 6. Consider

$$A = \begin{bmatrix} \lambda & 1 & 0 \\ 0 & \mu & 1 \\ 0 & 0 & \nu \end{bmatrix}.$$

For which λ, μ, ν is the matrix A diagonalizable?